IOT BASED SENTRY ALERT & MONITORING SYSTEM

Real-Time Motion & Health Monitoring using ESP8266

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Introduction

- Soldiers in remote or isolated regions often lack real-time health/motion monitoring.
- Emergency conditions like fainting or intrusions go undetected.
- I propose a Wi-Fi-based low-power system using IoT to detect and alert in real time.

OBJECTIVES:

Monitor vitals (BPM, motion) using wearable sensors

Detect motion/fainting events and alert instantly.

Display data on OLED & live dashboard and Use Wi-Fi for real-time updates and alerts

PROBLEM STATEMENT:

No continuous monitoring of vitals or motion

Delayed response to accidents or intrusions

Need for wireless, real-time, low-power alert system.

HARDWARE COMPONENTS:

D1 Mini ESP8266 – Main controller with Wi-Fi

MAX30102 – Pulse & SpO_2 sensor

MPU6050 – Accelerometer + Gyroscope

HC-SR501 – PIR Motion Sensor

OLED Display – Alerts and status

Buzzers, Battery, JST, Mini 360 Buck Converter

PROJECT ARCHITECTURE:

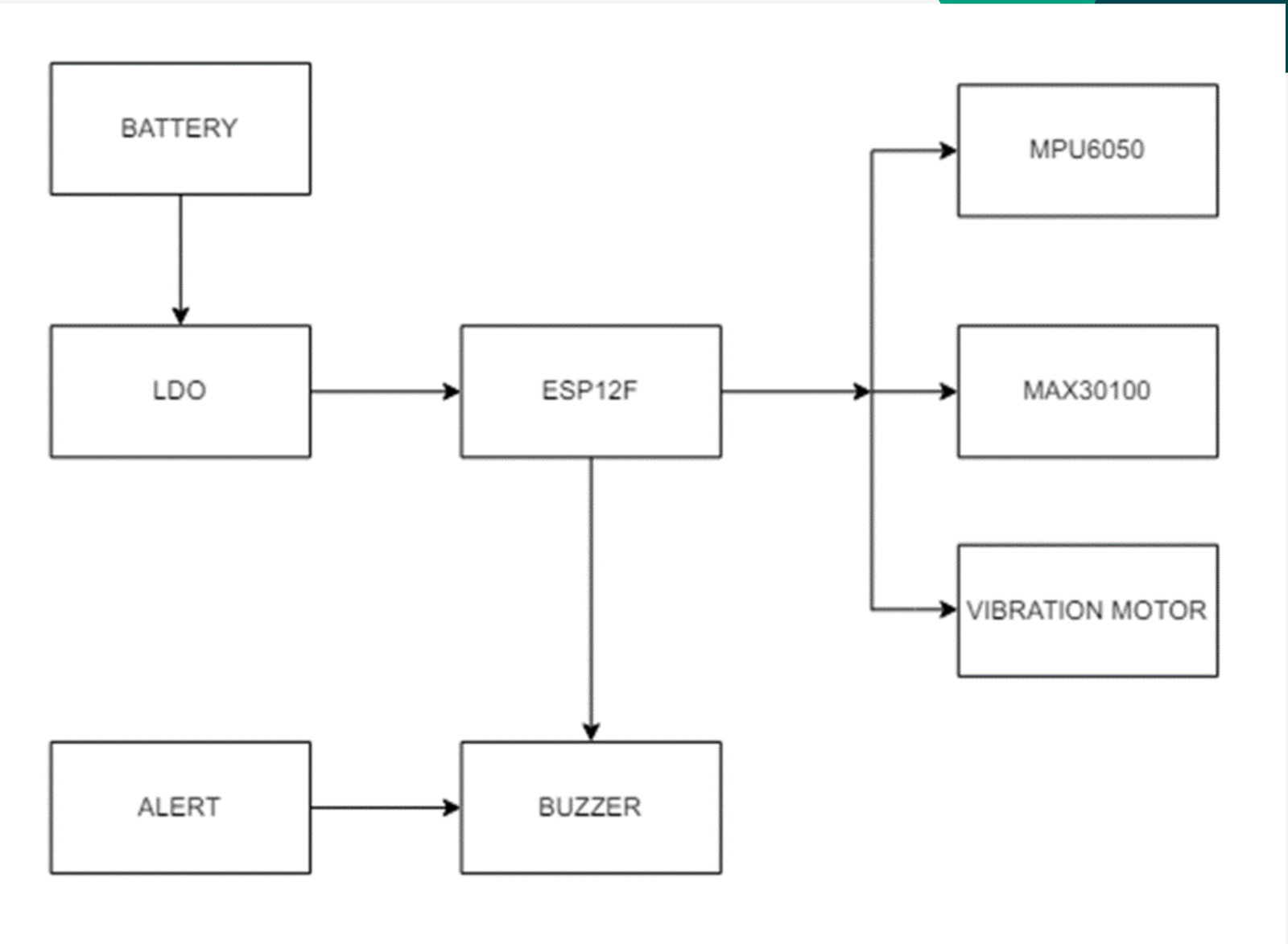
Fig 1 :Motion Node

Motion Node → Dashboard (HTTP):

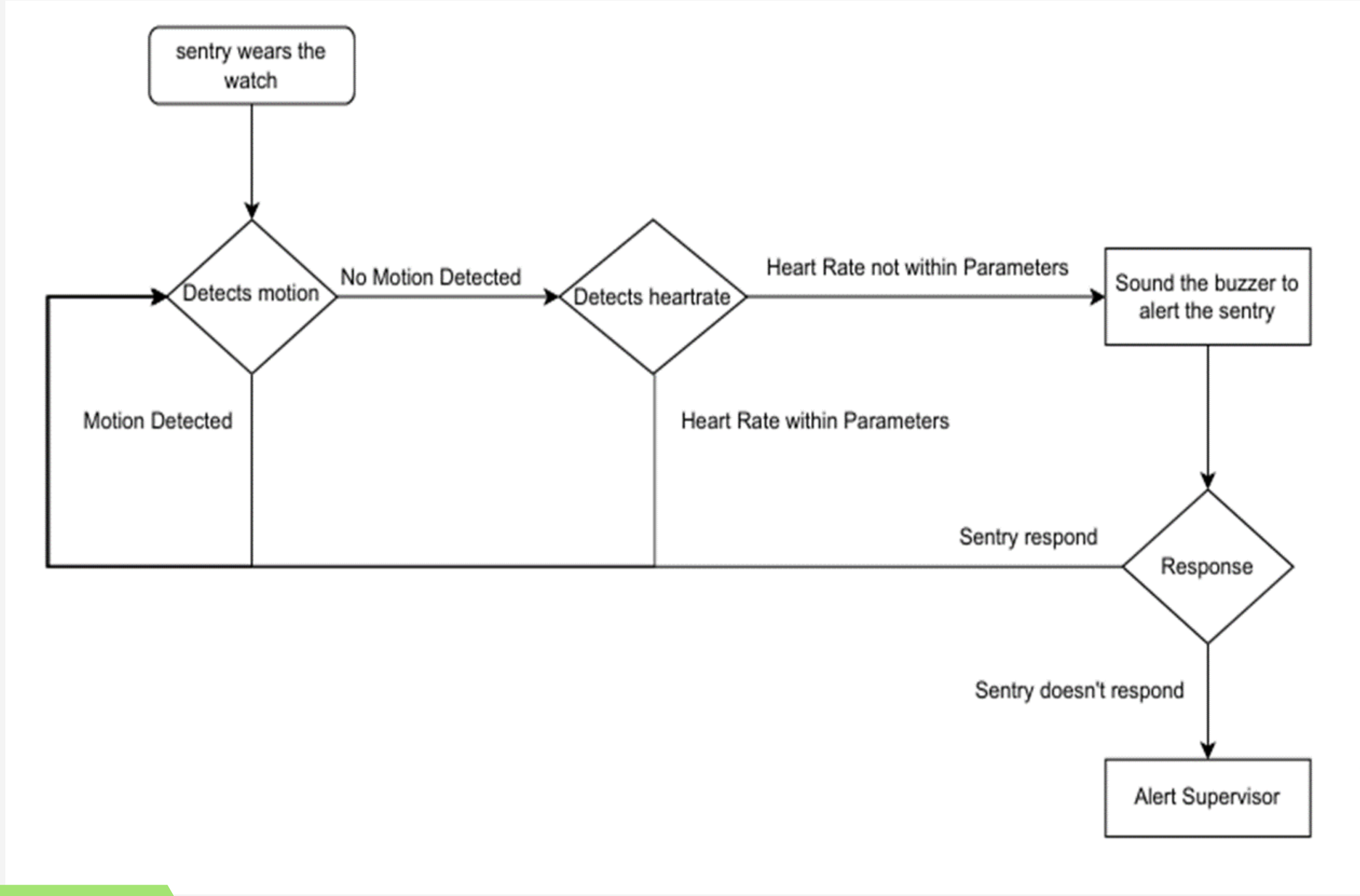
Watch Node → Dashboard (Vitals via HTTP)

Dashboard → OLED & Buzzer ESP (WebSocket)

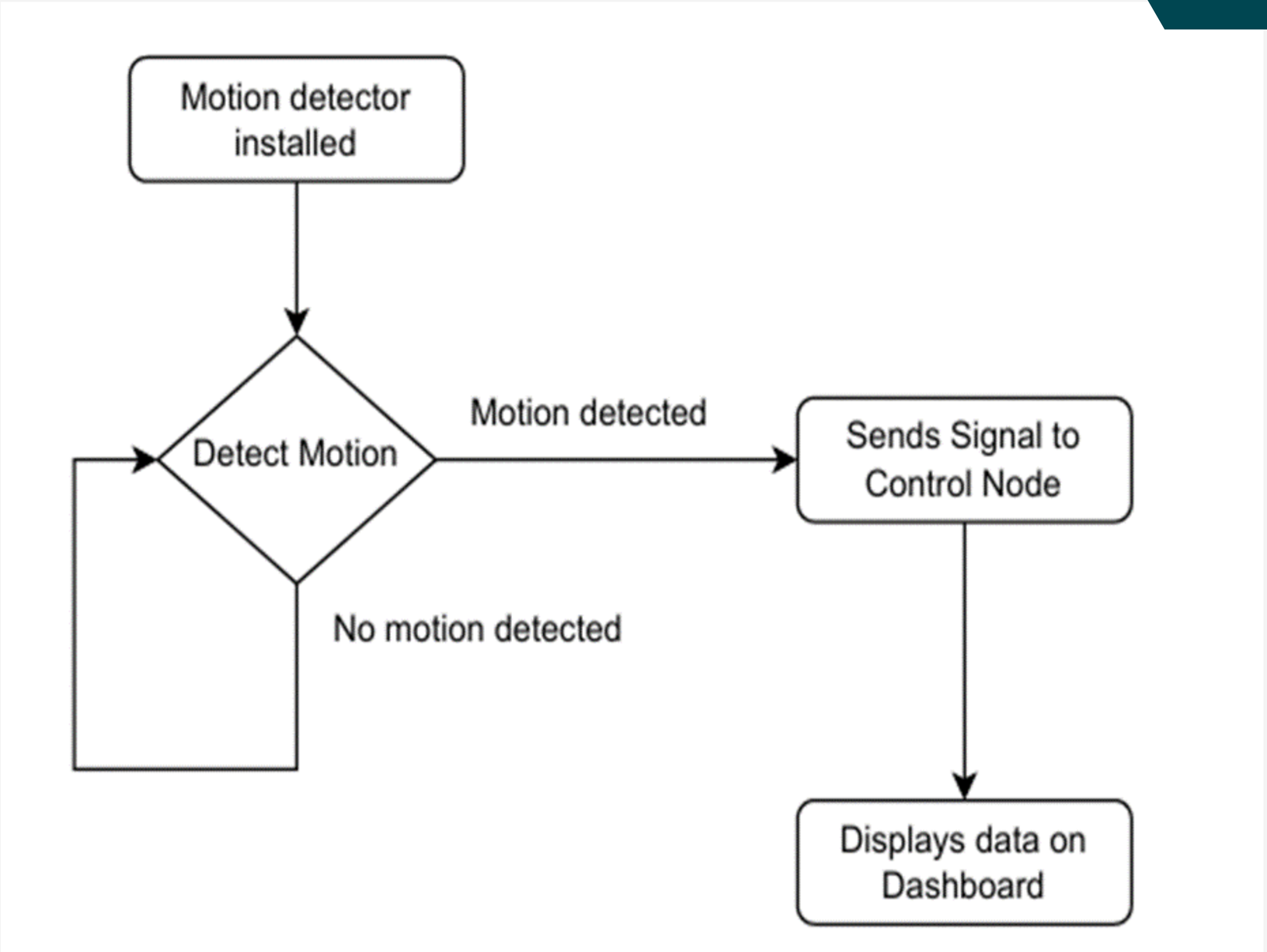
BLOCK DIAGRAM OF SENTRY ACTIVITY BAND



FLOW DIAGRAM OF SENTRY ACTIVITY BAND



FLOW DIAGRAM OF MOTION NODE



KEY FEATURES:

- Real-time monitoring & WebSocket dashboard
- OLED + buzzer-based alerts
- Compact & low power (deep sleep supported)
- Multi-node support for scalability
- Fault tolerance (reconnect on Wi-Fi loss)

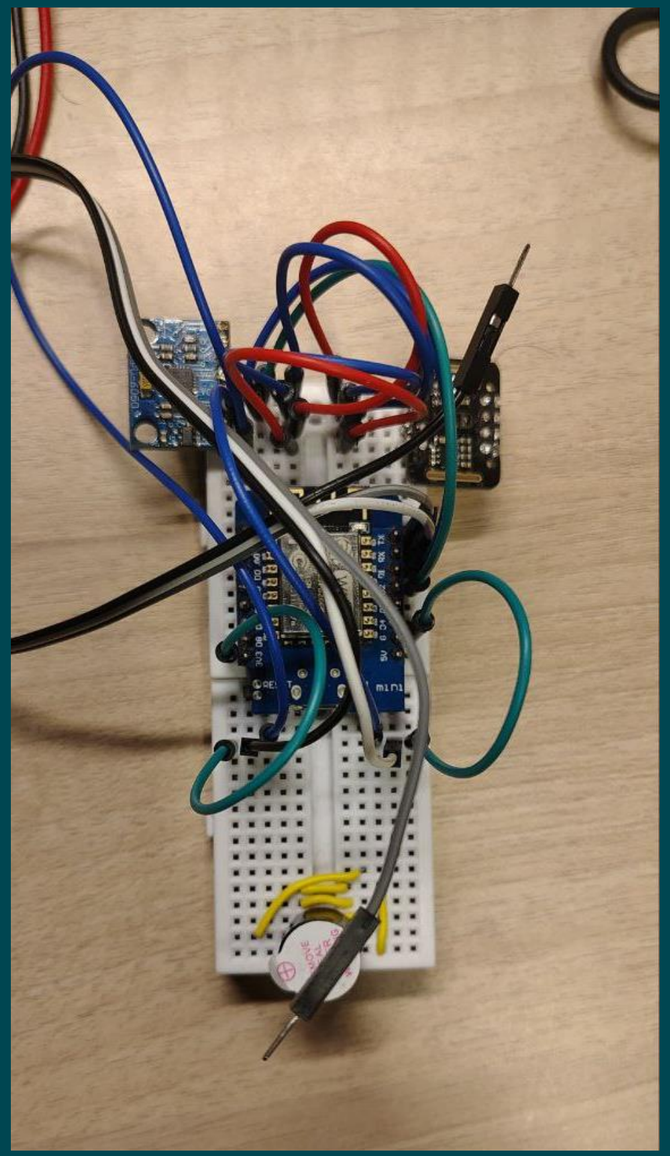


Fig 2: Watch Node

COMMUNICATION STANDARDS

IEEE 802.11

Enables reliable Wi-Fi communication between all ESPs

IEEE 2700-2017

Ensures sensor accuracy in wearable systems

I2C

Efficient sensor communication

HTTP + WebSocket

Fast and asynchronous data transfer

References & Libraries

Libraries Used:

- ESPAsyncWebServer, WebSocketsServer, Wire.h
- Adafruit_SSD1306, Adafruit_GFX, MAX3010x, MPU6050

References:

[1] IEEE Standard 802.11, Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specs, IEEE, 2016. [Online]. Available: https://standards.ieee.org/802-11.
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[3] Espressif Systems, “ESP8266EX Datasheet,” [Online]. Available: https://www.espressif.com. Accessed: Feb. 26, 2025, 9:22 AM.

THANK YOU